Sensitivity of Multifrequency Polarimetric SAR Data to Postfire Permafrost Changes and Recovery Processes in Arctic Tundra

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***Abstract*— We used full-polarimetric L-band and P-band synthetic aperture radar (SAR) data collected from the recent NASA Arctic Boreal Vulnerability Experiment (ABoVE) airborne campaign and Sentinel-1 C-band dual-polarization data to understand the sensitivity of radar backscatter intensity and phase to fire-induced changes in the surface and subsurface soil processes in Arctic tundra underlain by permafrost. The 2007 Anaktuvuk River fire on the Alaska North Slope was used as a case study. At ∼10-year postfire, we observed a strong increase (>∼3–4 dB) in the low-frequency radar backscatter in severely burned areas during the thaw season, in contrast to limited (<∼0.5 dB) C-band backscatter differences (VV and VH) between burned and unburned areas. However, C-band winter backscatter is generally higher (>1 dB) in burned areas than the adjacent unburned areas. Polarimetric decomposition analysis indicated a general trend toward more random surface scattering, and strong increases in double-bounce scattering and volume scattering power at both P- and L-band in the burned areas. The ice-rich yedoma region shows the largest backscatter increases in burned areas and the highest correlation with burn severity and microtopography changes. The above backscatter changes are attributed to increasing surface roughness and microtopography due to ice-wedge degradation and thermokarst development and increasing subsurface scattering due to an overall drier and deeper active layer in burned areas. Among all frequencies, P-band shows consistently larger contrast in backscatter power and phase between burned and unburned areas, which makes it potentially more useful to study fire–permafrost interactions in the Arctic over decadal time scales.**

***Index Terms*— P-band, permafrost, polarimetric decomposition, tundra fire.**

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I. Introduction

FIRE is a major disturbance in the boreal and Arctic region, with large impacts on vegetation and regional carbon cycle, soil hydrology, and permafrost conditions [1]–[3]. Tundra fire can trigger and promote permafrost degradation by removing the insulating surface vegetation and organic layer, altering the surface albedo and energy budget, and promoting ice-wedge degradation and thermokarst development [4]–[6]. Fire can also release a large amount of carbon emissions to the atmosphere through the combustion of vegetation, surface, and soil organic matter, potentially offsetting Arctic greening [1]. Therefore, potential climate-driven increases in tundra fire disturbance may represent a positive feedback intensifying the rate of global warming [7].

Satellite and airborne remote sensing have been widely used to map burned area, fire severity, and recovery. Optical spectral indices, including measures of vegetation greenness and the differential normalized burn ratio (dNBR), have been traditionally used for monitoring fire severity and postfire vegetation recovery [8]–[10]. However, a major limitation of optical sensors in fire monitoring is the lack of coverage during active fire events and the postfire recovery period due to cloud and aerosol contamination, especially in the northern high latitudes. In addition, optical remote sensing is more sensitive to surface vegetation greenness, which generally shows more rapid recovery than vegetation structure, biomass, and subsurface processes [11]. For example, in the boreal and Arctic region, the removal of the surface vegetation and organic layer can disrupt the equilibrium state of the underlying soil active layer, promote permafrost degradation and talik formation, and alter the surface and subsurface drainage pattern, which can take many years to recover to the prefire state [3], [12]. Alternatively, radar remote sensing can penetrate clouds and aerosols and is sensitive to postfire changes in vegetation biomass and structure, soil moisture, and ground subsidence,

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which provides additional useful information on postfire recovery [4], [10], [13], [14].

Fire removes surface vegetation and leaves the ground and/or tree trunks exposed, which can result in changes in both radar backscatter intensity and phase. Previous studies have generally reported a postfire increase in radar backscatter relative to prefire conditions, mainly due to surface vegetation removal and associated increases in surface roughness and scattering power; however, the resulting backscatter changes also depend on surface moisture conditions, as well as the radar frequency and viewing geometry [10], [15]–[17]. Higher microwave frequencies (e.g., C-band and X-band) generally show relatively rapid recovery within a few years following fire due to their strong sensitivity to surface vegetation conditions, whereas lower frequencies (L- and P-bands) generally show a slower recovery partly due to their stronger sensitivity to understory conditions and changes in a postfire vegetation structure [18]–[20].

Full-polarimetric synthetic aperture radar (PolSAR) data have advantages over single- or dual-polarization (Pol) data in its ability to provide a more complete and accurate description of the scattering processes. Using polarimetric target decomposition, we can effectively separate contributions from specific scattering mechanisms, allowing for improved retrieval of biophysical parameters of interest [21]. Many studies have used PolSAR data to map fire perimeters and burn severity and characterize postfire vegetation recovery through assessing the prefire and postfire changes in the radar scattering components and other polarimetric parameters [21]–[23]. Different scattering mechanisms show varying sensitivity to burn severity, depending on the amount of vegetation canopy damage, radar frequency, topography, and local incidence angle variations. Using full-polarimetric data generally allows for more accurate characterization of fire perimeters and burn severity with respect to backscatter intensities, with longer wavelengths such as L-band showing higher sensitivity to fire disturbance effects, especially in the forest [18]. However, we are only aware of a single study that examined the P-band PolSAR response to postfire recovery [24], and few studies have compared its sensitivity with PolSAR data at other frequencies, particularly in the Boreal–Arctic region, where fire–permafrost interactions play an important role in the postfire biophysical recovery.

Extensive coordinated airborne L-band (1.26 GHz) and P-band (0.43 GHz) radar data were collected during the recent NASA Arctic Boreal Vulnerability Experiment (ABoVE) campaigns, which offers a unique opportunity to study the potential of long-wavelength PolSAR data (especially P-band) in characterizing the fire and permafrost interaction and potential impact on postfire recovery processes [25]. In this study, we used the 2007 Anaktuvuk River tundra fire, occurred on the Alaska North Slope, as an example, to evaluate the sensitivity of multiple P- and L-band polarimetric parameters to changes in surface and subsurface processes after ten-year postfire. Besides the airborne data, we also included the Sentinel-1 C-band data to evaluate how multitemporal radar backscatter responds to postfire permafrost changes and recovery processes. Our analysis was guided by the following hypotheses. First, fire-induced thermokarst and ice-wedge degradation in the permafrost area will lead to increase in surface roughness, which may enhance the backscatter at all frequencies. In addition, fire removal or reduction of surface organic layer will accelerate permafrost and active-layer thawing. At ten-year postfire, both the active-layer deepening and thermokarst occurrence will very likely lead to surface drying due to enhanced soil drainage, which may result in different polarimetric responses, especially at the longer wavelengths. Here, we mainly used polarimetric decomposition and correlation analysis to investigate the potential response of polarimetric parameters and backscatter to the above processes.

II. Data and Methods

A. Study Area and in Situ Data

We conducted a case study of the 2007 Anaktuvuk River fire (ARF) scar on the Alaska North Slope (Fig. 1). This fire started in July and lasted through September 2007, burning an area of $\sim$1039 km^2 of Arctic tundra and representing the largest fire on record for the tundra biome [8]. More than 80% of the ARF was burned at moderate to extreme severity, and the fire released $\sim$2.1 Tg of carbon into the atmosphere. Fire consumed $\sim$6.1 cm of the surface organic layer on average, with the mean soil organic layer depth reduced from 21.5-cm prefire to 15.4-cm postfire [1]. Recent field and remote sensing-based studies observed substantial ground subsidence and development of deep thaw cracks, mainly driven by postfire ice degradation and thermokarst formation [5], [6], [26] (Fig. 2). The tundra appears to initially become wetter postfire and then drier as water drains through degrading ice troughs and a deepening active layer [26].

The NASA ABoVE campaign includes the detailed field measurements coincident with the airborne SAR flight campaign across the regional domain. The field data collections include extensive thaw depth and volumetric soil moisture (VSM) measurements to validate remotely sensed active layer thickness (ALT) and soil moisture [27]. The soil moisture data were collected using HydroSense probes, ground penetrating radar (GPR), and DualEM sensor. GPR and DualEM provide soil moisture measurements for the entire active layer, while the HydroSense probes provide soil moisture measurements averaged at two different soil layers (0–12- and 0–20-cm depths). Thaw depth measurements were mainly conducted using mechanical probing, with a small number of measurements from GPR retrievals. At our study sites (Fig. 1(a) and Table I), VSM was available from the GPR and HydroSense measurements, and thaw depth was measured by mechanical probing. There are multiple observations at each site, which were collected from a 100 $\times$ 40 m grid and four 30-m transects emanated from the grid center (Table SI) [28]. The *in situ* data collected in mid-August 2017 show an overall deeper thaw depth with drier condition in both the top layers (0–12 and 0–20 cm) and the entire active layer at severely burned sites comparing with unburned or moderately burned sites (Fig. 3), which is consistent with warmer soil temperatures recorded in the burned areas [26]. The thaw depth data in late-August

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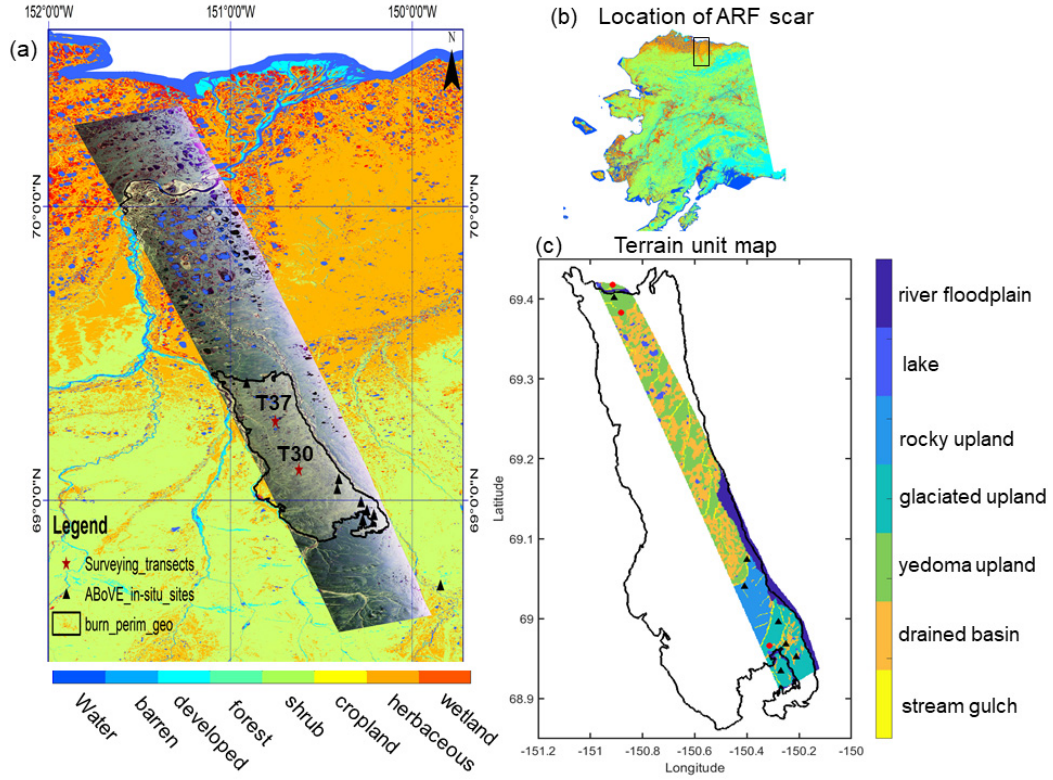


Fig. 1. (a) 2007 ARF scar and (b) its locations in Alaska shown as the black rectangle. (c) Terrain unit map derived from a recent LiDAR survey [5] within the fire perimeter. The black triangles (a) and (c) Location of *in situ* sites during the ABoVE field campaign (Table I). Three additional sites (red dots) representing different landforms and burn severity were also included for backscatter comparison (Table SII). In (a), the Pauli image from one of the L-band flight lines was overlain on the 30-m NLCD land cover map, with two dominant tundra types in the study area, including herbaceous tundra in the north and shrub tundra in the south. Surface open water and wetlands are also common in the north.

TABLE I

In Situ SITES WITHIN THE STUDY DOMAIN ACQUIRED FROM THE ABOVE FIELD CAMPAIGN

Coordinates	Burn severity	Landform	Observation year	Observables
68.93°N, 150.27°W	Unburned	Glaciated upland	2017, 2018	VSM, ALT
68.95°N, 150.21°W	Moderate burn	Glaciated upland	2017, 2018	VSM, ALT
69.00°N, 150.28°W	Severe burn	Glaciated upland	2017, 2018	VSM, ALT
68.97°N, 150.25°W	Thermokarst (severe burn)[1]	Glaciated upland	2017	VSM, ALT
69.40°N, 150.91°W	Low burn	Yedoma	2017	VSM, ALT
69.04°N,150.41°W	Severe burn[2]	Rocky upland	2018	VSM, ALT

[1]determined by Landsat-based dNBR = 806±24; [2]Field survey indicates it is an unburned site, while the dNBR map shows high burn severity (835±49)

2018 show a similar pattern as in 2017, while the VSM data vary for sites with different burn severity and landforms (Fig. S1). There are no meteorological stations that represent the entirety of the study area; therefore, we extracted spatially interpolated daily meteorological data from the Daymet dataset [29] to examine local climate variations at the study sites.

B. SAR Data Processing

PolSAR data at multiple frequencies, including C-, L-, and P-bands (Table II), were used to study the postfire tundra recovery in the ARF region. Sentinel-1 satellite C-band Ground Range Detected (GRD) scenes (path 94 and frame 225) in 2017 and 2018 were downloaded from the NASA SAR data archive at the Alaska Satellite Facility. The GRD data were processed using the SNAP-Python package v7.0. There are 24 and 21 images in total at the VV and VH channel acquired in years 2017 and 2018, respectively. There were no GRD data released for the granule of path 94 and frame 225 during April and May, 2018; other granules were not used due to different viewing geometries of the images.

TABLE II

INFORMATION ON THE SAR IMAGES USED IN THE STUDY

Frequency	Sensor	Polarization	Dates
C-band (5.4 GHz)	Sentinel-1	VV, VH	24 images for 2017 (Path 94); 21 images for 2018 (Path 94)
L-band (1.26 GHz)	NASA Airborne L-band SAR	Full	2017-06-21 (no full coverage for anaktE flight line), 2017-09-16, 2018-08-26
P-band (0.43 GHz)	NASA Airborne P-band SAR	Full	2017-06-06, 2017-08-13

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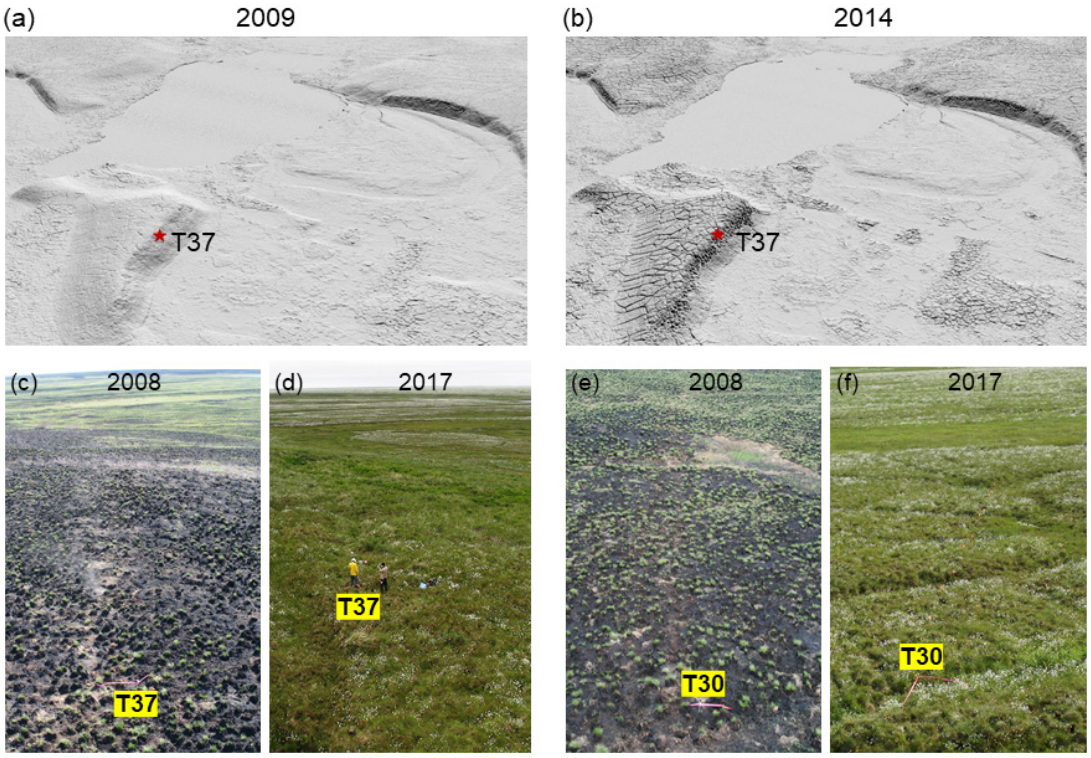


Fig. 2. Increasing surface roughness during late postfire recovery due to ice-wedge degradation, as shown in (a) and (b) hillslope images at one surveying transect (T37) and (c)–(f) two pairs of aerial photographs at T37 and another transect (T30) [Fig. 1(a)]. The hillslope images were created using 2009 and 2014 LiDAR DTM data [5] and shown with a 3× vertical exaggeration. The labels in the aerial photographs show the start point of the surveying transect.

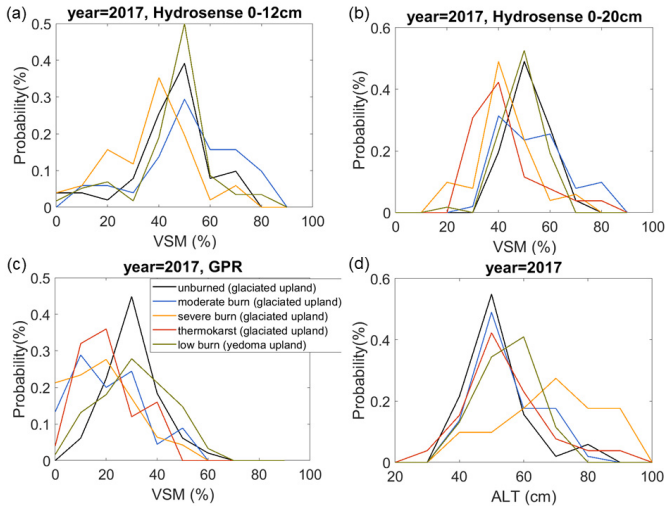


Fig. 3. Comparison of the distributions in *in situ* VSM (%) at (a)–(c) different soil depths and (d) ALT between burned and unburned tundra sites (Table I), collected on August 18 and 19, 2017.

The Sentinel-1 GRD data processing includes thermal and radiometric calibration, speckle noise filtering, and geolocation calibration using the 5-m Alaska IFSAR DEM data. The output images were resampled to 40-m resolution. The local incidence angle of the SAR images mainly ranges from ∼34° to 39° over the ARF region.

Airborne L- and P-band SAR data were obtained during ABoVE airborne campaigns in 2017 (L- and P-bands) and 2018 (L-band only) using the NASA/JPL's Uninhabited Aerial Vehicle SAR (UAVSAR) system flying aboard NASA's C-20 or G-III aircraft [25] (Table II). Both L- and P-band observations are fully polarimetric and available at three independent polarization channels (HH, VV, and VH). The 2017 acquisitions of L- and P-band data did not temporally coincide, though they both cover the early (June) and late (August and September) thaw periods. Two adjacent flight lines at West (anaktW) and East (anaktE) were acquired at both frequencies to cover the entire fire scar. The L-band incidence angle ranged from 20° to 70°, while the narrower P-band swath covered incidence angles from 20° to 60° in June 2017 and from 20° to 55° in August 2017. The spatial resolution of the L-band data is ∼6 m (0.2 arc-second), and UAVSAR instrument has a noise-equivalent sigma zero (NESZ) of ∼ −47 dB depending on the incidence angle. The spatial resolution of the P-band data is ∼15 m (0.5 arc-second), and the AirMOSS P-band instrument has an NESZ of −40 dB [30]. Observations at both frequencies were radiometrically and polarimetrically calibrated by the project team at NASA JPL (https://uavsar.jpl.nasa.gov/cgi-bin/data.pl) and provided as Stokes parameters. In this study, the multilook SAR data, with a spatial resolution of ∼60 m (2 arc-second) at L-band and ∼90 m at the P-band (3 arc-second), were used in order to reduce the speckle noise in the SAR data.

Incidence angle has a large impact on the radar backscatter response. We examined how the radar backscatter varies with incidence angle using the data in the overlapping areas of the two radar swaths (anaktE and anaktW). The L-band flights have overlapping areas covering most of the fire scar due to a

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wider processed swath except for the June 2017 acquisition, which only acquired data in the area south of 69.8°N for the anaktE flight line. The average incidence angle of the two L-band flight lines ranges from 48° to 58° for the overlapping areas (Fig. S2). The two P-band flights have limited overlapping areas for the June acquisition and no overlapping areas for the August acquisition. Based on Lambert's law, the radar backscatter can be related to the cosine square of the incidence angle; therefore, previous studies have used cosine correction to normalize the radar response (σ°_{θ}) from any incidence angle (θ) to a reference incidence angle (θ_{ref})

$$\sigma^{\circ}_{\text{ref}} = \frac{\sigma^{\circ}_{\theta}\cos^{n}(\theta_{\text{ref}})}{\cos^{n}(\theta)} \text{ or}$$
$$10\log_{10}(\sigma^{\circ}_{\text{ref}}) = 10\log_{10}(\sigma^{\circ}_{\theta}) + n(\cos(\theta_{\text{ref}}) - \cos(\theta)) \quad (1)$$

where n generally varies with surface roughness and for different land cover types and can be determined using a linear fit between backscatter observations in dB scale and cosine of incidence angle [31]. Previous studies have shown that one may obtain a reasonable fit between backscatter (at dB scale) and incidence angle with a linear approximation of the cosine function for incidence angle between 30° and 60° [32], [33].

In the ARF study area, we also observed a strong linear relationship ($r > 0.9$ and $p < 0.1$) between incidence angle changes and scattering power differences between the two flight line observations. The study area is dominated by herbaceous and shrub tundra [Fig. 1(a)], and we found negligible differences in the regression coefficients between the two tundra types. Therefore, we used a uniform linear regression for the two tundra types to perform the incidence angle normalization for each scattering power variable. For the analysis, we chose 50° as the reference incidence angle, which is close to the incidence angle of the acquisitions in the middle of the fire scar. However, we used 40° as the reference angle when comparing the C- and L-band backscatter changes between burned and unburned sites (Fig. 4) since this value is close to the C-band incidence angle in the study area. The normalization was only conducted for the overlapping areas in order to reduce the bias in areas with very low incidence angle. We also observed a strong linear relationship between the P-band backscatter and the incidence angle changes; however, we did not normalize the P-band data due to a narrow range of incidence angle of the observations in the overlapping areas. Instead, we combined the observations from the two flights to generate a single set of observations. For the overlapping pixels, we selected observations with incidence angles close to 55° (the incidence angle in the middle of the two flight lines).

A quantitative understanding of the scattering mechanisms is required to interpret the L- and P-band SAR data and to serve as a guide for further development of radar-based permafrost and soil moisture retrieval algorithms [34]. Two main approaches have been developed to decompose the fully PolSAR observations, including eigen- and model-based decomposition methods. The eigen-based decomposition is based on the interpretation of the eigenvalues and eigenvectors of the coherency matrix and requires no preassumption on the type of scattering components [35]. In contrast, the model-based approach decomposes the total backscattering signal into the component contributions from different scattering processes based on predefined scattering matrixes for each scattering mechanism [36]. The scattering coherency matrix ($[T]$) contains important information on the scattering matrix and is widely used to extract polarimetric parameters on distributed scatterers (2), as shown at the bottom of the page, where S indicates the scattering response and the superscript * represents the complex conjugate. The diagonal elements of the coherency matrix can be interpreted as the backscattered power for different scattering mechanisms, and the off-diagonal elements contain information on the phase difference of scattering mechanisms. For a natural surface with reflection

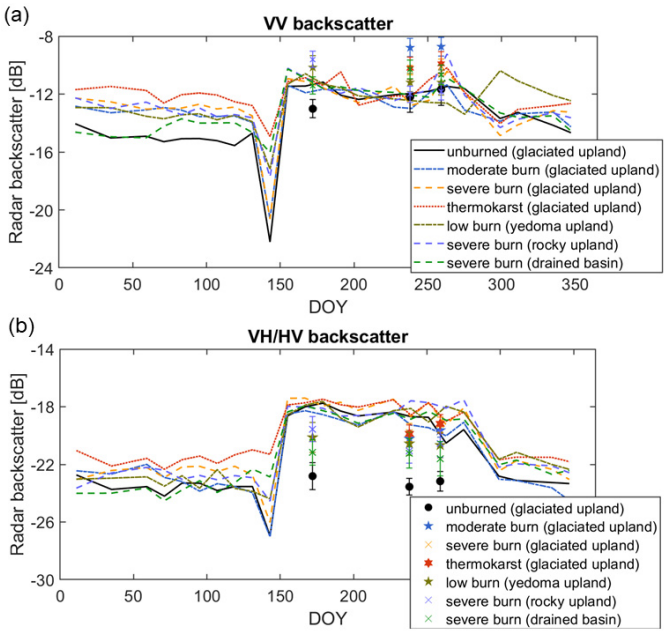


Fig. 4. Time series of (a) C-band VV and (b) C-band VH radar backscatter (represented by solid and dashed lines) at the ABoVE *in situ* sites in 2017. The L-band backscatter data were available for limited dates and, as a comparison, shown as symbols with the same color as the C-band backscatter lines for different sites. All data were extracted and averaged for a spatial window of ~240 m × 240 m overlaying each site. The incidence angle at C-band ranges from 33° to 37.5°, while the L-band backscatter was normalized to a 40° incidence angle.

$$T = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{12}^{*} & T_{22} & T_{23} \\ T_{13}^{*} & T_{23}^{*} & T_{33} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} \langle|S_{\text{HH}} + S_{\text{VV}}|^2\rangle & \langle(S_{\text{HH}} + S_{\text{VV}})(S_{\text{HH}} - S_{\text{VV}})^*\rangle & 2\langle S_{\text{HV}}^*(S_{\text{HH}} + S_{\text{VV}})\rangle \\ \langle(S_{\text{HH}} - S_{\text{VV}})(S_{\text{HH}} + S_{\text{VV}})^*\rangle & \langle|S_{\text{HH}} - S_{\text{VV}}|^2\rangle & 2\langle S_{\text{HV}}^*(S_{\text{HH}} - S_{\text{VV}})\rangle \\ 2\langle S_{\text{HV}}(S_{\text{HH}} + S_{\text{VV}})^*\rangle & 2\langle S_{\text{HV}}(S_{\text{HH}} - S_{\text{VV}})^*\rangle & 4\langle|S_{\text{HV}}|^2\rangle \end{bmatrix} \quad (2)$$

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TABLE III
POLARIMETRIC PARAMETERS ANALYZED IN THE STUDY

Polarimetric decomposition/scattering matrix	Variables	References
Covariance matrix [C]	HH, HV and VV backscatter (C_{11}, C_{22}, C_{33}), C_{13} amplitude and phase (HH/VV ratio and phase difference)	N/A
Coherency matrix [T]	T_{11} ($\lvert S_{HH}+S_{VV}\rvert$), T_{22} ($\lvert S_{HH}-S_{VV}\rvert$), |), T_{33} ($\lvert S_{HV}\rvert$, equivalent to C_{22}), T_{12} phase ($\lvert S_{HH}+S_{VV}\rvert$ and $\lvert S_{HH}-S_{VV}\rvert$ phase difference)	N/A
Entropy-Alpha decomposition	alpha, entropy	[35]
Freeman-Durden 3 component decomposition	Surface (Freeman_Odd), volume (Freeman_Vol) and double bounce (Freeman_Dbl) scattering power	[36]
Touzi decomposition	Dominant and second scattering type phase (TSVM_phi_s1, TSVM_phi_s2)	[37]

symmetric scattering, the cross products of scattering power of cross-Pol and co-Pol channels (e.g., $S_{\mathrm{hv}}S_{\mathrm{vv}}^{*}$ and $S_{\mathrm{hh}}S_{\mathrm{hv}}^{*}$) generally have much lower power than that of the co-Pol or cross-Pol observations; therefore, T_{13} and T_{23} can be ignored. Hence, we only included the three diagonal elements and the off-diagonal T_{12} phase in our analysis (Table III). Besides the coherency matrix, covariance matrix ([C]) is also widely used, which allows for the analysis of single-polarization channels as well as their relations

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{12}^{*} & C_{22} & C_{23} \\ C_{13}^{*} & C_{23}^{*} & C_{33} \end{bmatrix} = \begin{bmatrix} \langle |S_{\mathrm{HH}}|^2 \rangle & \sqrt{2}\langle S_{\mathrm{HH}} S_{\mathrm{HV}}^{*} \rangle & \langle S_{\mathrm{HH}} S_{\mathrm{VV}}^{*} \rangle \\ \sqrt{2}\langle S_{\mathrm{HV}} S_{\mathrm{HH}}^{*} \rangle & 2\langle |S_{\mathrm{HV}}|^2 \rangle & \sqrt{2}\langle S_{\mathrm{HV}} S_{\mathrm{HV}}^{*} \rangle \\ \langle S_{\mathrm{VV}} S_{\mathrm{HH}}^{*} \rangle & \sqrt{2}\langle S_{\mathrm{VV}} S_{\mathrm{HV}}^{*} \rangle & \langle |S_{\mathrm{VV}}|^2 \rangle \end{bmatrix}. \quad (3)$$

Here, we included the diagonal elements of [C] matrix representing the directly backscattering coefficients for HH, HV, and VV polarizations and the C_{13} amplitude and phase representing the ratio and phase difference of HH and VV channels (Table III). The Strokes parameters provided by the UAVSAR project team were input into the PolSARpro (v6.0) software to generate coherence and covariance matrices, which were then used as the inputs for the target decomposition.

We included the outputs from three popular decomposition methods in our analysis [35]–[37] (Table III). The first two methods are based on eigenvector analysis, and the Freeman–Durden decomposition is model-based. We included two polarimetric parameters from the Cloude–Pottier decomposition, including the alpha angle representing the dominant scattering mechanism and the entropy representing the randomness of the scattering. The Touzi decomposition is based on a similar eigenvector decomposition as the Cloude–Pottier method; however, the parameters are phase-invariant. The dominant and second scattering phases, representing the phase difference between the vector components of the trihedral-dihedral bias, have been shown to be sensitive to subsurface hydrology changes in northern wetlands [37] and were included in our study. Other parameters, including the scattering and orientation angle and helicity, either showed a similar pattern as the above parameters or random distribution in our study area and thus were not included. Finally, the Freeman–Durden method decomposes the scattering matrix into three components, including odd-bounce (surface) scattering, double-bounce scattering, and volume scattering, and we included the power intensity of the three scattering mechanisms in our analysis.

We performed the Pearson correlation analysis between the polarimetric parameters and a Landsat-based dNBR map [8] to investigate how burn severity affects the radar scattering process at different frequencies. Postfire recovery in the tundra area differs greatly in different landforms [5], [26]; therefore, we also examined the radar response for different landforms in the 2009/2014 repeat light detection and ranging (LiDAR) survey region [5] shown in Fig. 1(c). The ABoVE *in situ* measurement sites (Fig. 1(c) and Table I) cover the major terrain units in the LiDAR swath; however, the severe and moderate burned sites in the glaciated upland area are located outside of the overlapping area of the two L-band flights. Therefore, we also included three additional sites with similar land cover, burn severity ratio, and slope representing glaciated and yedoma uplands (Table SII). For the site comparison, we extracted the radar backscatter using a window size of ~240 m × 240 m.

III. RESULTS

A. Backscatter Analysis

Our results showed limited (~<1 dB) differences in VV and VH backscatter at C-band between burned and unburned sites during the growing season at ten-year postfire within the ARF domain (Fig. 4). However, the burned areas generally show higher radar backscatter intensity at all channels for both P- and L-bands. The backscatter difference at L-band can reach up to 4 dB between severely burned and unburned areas at the 40° incidence angle. The P-band backscatter also shows large differences between the burned and unburned areas; however, the data are not included here due to inconsistent incidence angle among the sites and limited spatial overlap of acquisitions enabling the backscatter normalization process.

Even though limited differences in the C-band backscatter between burned and unburned areas were observed during the summer, winter backscatter (January–April) showed consistent differences between burned and unburned (or low

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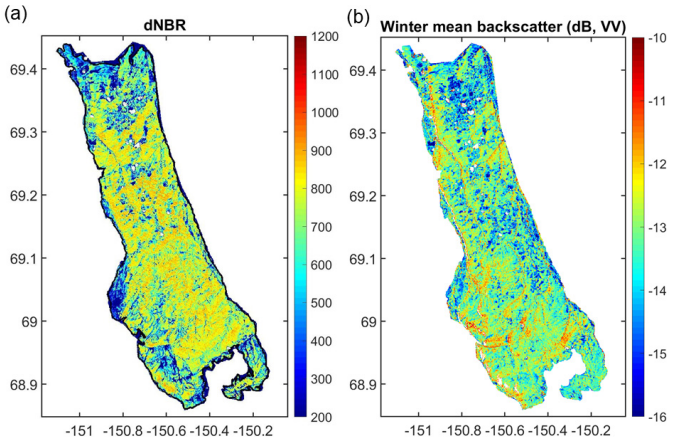


Fig. 5. Comparisons of the spatial pattern of burn severity indicated by (a) Landsat-based dNBR map and (b) C-band winter backscatter averaged from DOY 11 to 131 in 2017. The majority of the pixels have an incidence angle from 34° to 39° with a mean incidence angle of 36.8°.

TABLE IV

STATISTICS OF C-BAND VV BACKSCATTER DURING WINTER[1] FOR DIFFERENT LANDFORMS WITHIN THE LIDAR SWATH

	Glaciated upland	Rocky upland	Drained basin	Yedoma upland	River floodplain
Fractional coverage	11.3%	9.9%	28.7%	28.7%	5.9%
Backscatter difference[2]	1.4±0.7	1.3±0.6	1.4±0.8	1.8± 0.6	-0.5±1.2
R vs dNBR	0.46	0.23	0.36	0.49	-0.14

[1]averaged from DOY 11 to 131; [2]backscatter difference [dB] between severely burned (dNBR>700) and low burn or unburned areas (dNBR<100); stream gulch and lake areas (Fig.1c) have limited coverage (<3%) and are not included.

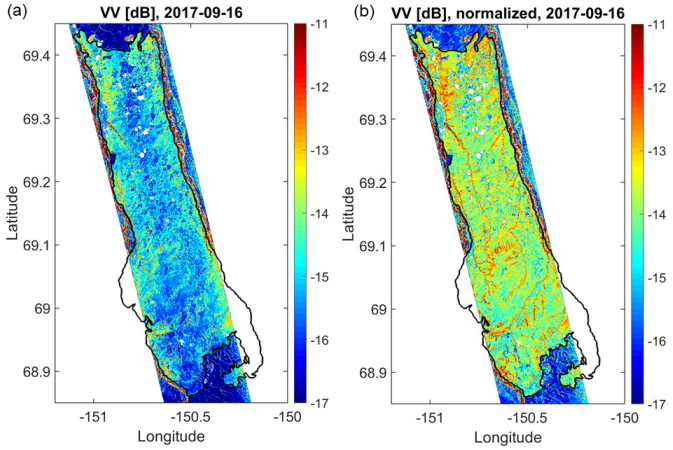


Fig. 6. Original map of L-band backscatter averaged using the observations from (a) two flight lines versus (b) normalized backscatter at 50° incidence angle. The black lines show the fire perimeter. Open water areas within the swaths were masked out using the NLCD land cover map.

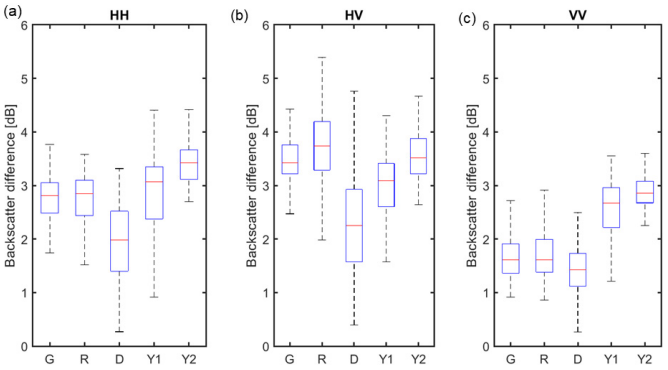


Fig. 7. Comparisons of L-band backscatter differences between unburned and burned sites representing different landforms at 50° incidence angle for the September 16, 2017, acquisition (G: Glaciated upland; R: Rocky upland; D: Drained basin; Y1: Yedoma low burn; Y2: Yedoma moderate burn). The central mark indicates the median, and the bottom and top edges of the box indicate the 25th and 75th percentiles, respectively. The backscatter was extracted within a spatial window of ∼300 m × 300 m over each site.

burn severity) regions (Figs. 4 and 5). Averaged over the fire scar, VV backscatter during winter (DOY 11–131) is consistently higher (1.2 ± 0.8 dB) at the severely burned areas (dNBR > 700) than unburned or low burn severity areas (dNBR < 100), while the backscatter is more variable during the summer (DOY 155–239) across the region, with a slightly lower backscatter on average (difference ≈ 0.3 dB) in the severely burned area than the low burn region. Winter VV backscatter also showed consistently higher correlation with dNBR ($R = 0.34$ and $p < 0.1$), while summer backscatter showed nonsignificant ($p > 0.1$) and more variable correlation with dNBR. However, there are also large differences in the winter backscatter among different landforms, with yedoma upland regions showing the highest correlation ($R = 0.49$ and $p < 0.1$) with dNBR and the largest backscatter differences between severely burned and unburned (or low burn severity) areas (1.8 ± 0.6 dB) (Table IV). River floodplain areas are different from all other landforms, with backscatter decrease with increasing burn severity (Fig. S3). The summer backscatter of different landforms is more variable due to vegetation growth and changes in surface soil moisture. Compared with VV backscatter, VH backscatter shows consistently lower correlation with dNBR.

Both the normalized and original L-band observations indicated overall higher backscatter in the burned areas than in the unburned areas (Figs. 6 and S4). The observations from the two flight lines were first averaged over the overlapping areas based on a strong linear relationship between backscatter and incidence angle; the composite image [Fig. 6(a)] has an incidence angle ranging from 48° to 58°, with a mean incidence angle of 53° (Fig. S2). The backscatter was then normalized to a 50° incidence angle using (1). The normalized backscatter image [Fig. 6(b)] showed a similar pattern, but with general increase in the backscatter especially in the middle of the image due to a lower incidence angle in the original composite image. Across the fire scar, VH shows the largest spatial variability, whereas VV shows the smallest variability.

L-band backscatter increases with burn severity at all acquisition dates for all landforms except for the river floodplain, with the largest increase in the yedoma and glaciated upland areas (Fig. S5). However, large variability occurs in the radar backscatter among different landforms and burn severity

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TABLE V
L-BAND BACKSCATTER DIFFERENCES BETWEEN BURNED AND UNBURNED SITES REPRESENTING DIFFERENT LANDFORMS[1]

	2017-09-16			2018-08-26		
	HH	HV	VV	HH	HV	VV
Glaciated upland	2.77±0.46	3.48±0.44	1.63±0.39	2.91±0.41	3.40±0.38	1.69±0.31
Rocky upland	2.73±0.50	3.80±1.00	1.73±0.51	2.81±0.44	3.48±0.97	1.49±0.48
Drained basin	1.90±0.78	2.31±1.17	1.41±0.52	2.49±0.72	2.84±0.96	2.04±0.48
Yedoma (mod burn)[2]	2.77±0.89	2.84±0.92	2.44±0.76	2.62±0.86	2.64±0.92	2.12±0.65
Yedoma (sev burn)[2]	3.41±0.36	3.57±0.50	2.86±0.33	3.31±0.32	3.38±0.42	2.51±0.31

[1]All backscatter values (units: dB) were normalized to 50° incidence angle and extracted for a window size of 300 × 300m over each site. The backscatter difference from the June 21, 2017 acquisition was not calculated due to a lack of coverage of the East flight line. [2]Burned sites include moderate (mod) and severe (sev) burns.

TABLE VI
CORRELATION COEFFICIENT BETWEEN BURN SEVEIRTY (DNBR) AND THE P-BAND SAR POLARIMETRIC PARAMETERS

	INC (40°-45°)		INC (45°-50°)		INC (50°-55°)	
	17-06-06	17-08-13	17-06-06	17-08-13	17-06-06	17-08-13
HH (C_{11})	0.56	0.52	0.53	0.54	0.55	0.51
HV (C_{22})	0.53	0.51	0.45	0.50	0.49	0.51
VV (C_{33})	0.52	0.38	0.43	0.38	0.45	0.35
HH/VV ratio (C_{13} amplitude)	0.28	0.47	0.35	0.50	0.34	0.54
HH-VV phase difference[1] (C_{13} phase)	0.58	0.57	0.54	0.52	0.52	0.52
T_{11} ($\|S_{HH}+S_{VV}\|$)	0.54	0.44	0.46	0.46	0.47	0.41
T_{22} ($\|S_{HH}-S_{VV}\|$)	0.55	0.52	0.50	0.53	0.55	0.52
T_{12} (Pauli) phase[1]	0.51	0.58	0.45	0.55	0.38	0.57
Freeman_Odd	0.56	0.45	0.47	0.46	0.49	0.52
Freeman_Dbl	0.53	0.49	0.52	0.52	0.56	0.54
Freeman_Vol	0.55	0.53	0.46	0.52	0.50	0.56
alpha	0.40	0.46	0.42	0.50	0.45	0.53
entropy	0.23	0.32	0.21	0.39	0.18	0.34
TSVM_phi_s1[1]	0.62	0.57	0.54	0.44	0.39	0.27
TSVM_phi_s2[1]	-0.62	-0.62	-0.54	-0.50	-0.49	-0.35

[1]indicates phase related variables. The correlation was calculated for each 5° INC bin. The August flight line contains very few pixels with INC from 55° to 60°, so this INC bin was excluded from the analysis.

zones. This can also be seen from the backscatter differences extracted at the ABoVE *in situ* sites as shown, in Figs. 7, S6, and S7. The backscatter differences were calculated between the burned and reference unburned sites for each landform (Tables I and SII). The unburned site at the glaciated upland region was also used as the reference site for the adjacent rocky upland and drained basin burned site, due to similar slope and land cover conditions among those sites. Larger differences (>∼2 dB) were observed in the HV and HH polarization backscatter between the burned and unburned areas than for the VV polarization. Among all landforms, the yedoma sites showed the largest differences in HH and VV backscatter between the burned and unburned areas (Table V). At the HV polarization, backscatter differences (>∼3 dB) are comparable among the yedoma, glaciated, and rocky uplands. The backscatter differences showed a slightly larger dynamic range among the different sites for the September 16, 2017, acquisition than for the August 26, 2018 acquisition, with the latter date having much lower accumulated precipitation in the prior week.

B. Polarimetric Analysis

Previous studies using dual-polarization SAR data indicated that the Cloude–Pottier decomposition method is not effective for separating different scattering mechanisms unless both VV and HH channels were included [38]. Therefore, in this section, we only included the decomposition results based on the full-polarimetric L- and P-band data.

1) P-Band: Among all elements of the coherency matrix, the T_{22} and T_{12} phases (i.e., the Pauli phase) are least affected

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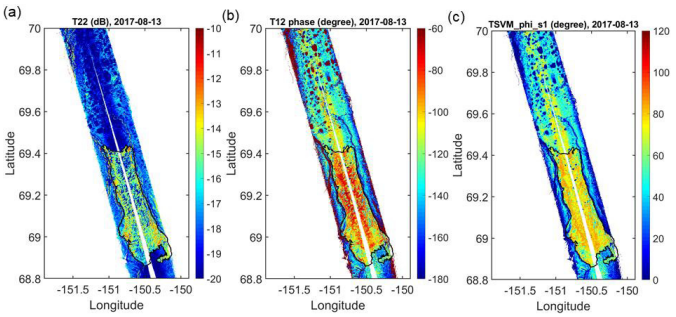


Fig. 8. Maps of the P-band coherency matrix elements (a) T_{22} (double-bounce scattering), (b) T_{12} (Pauli phase), and (c) Touzi dominant scattering-type phase over the ARF scar (black solid line) in August 13, 2017. All variables show large contrast between the burned and unburned areas. The incidence angle for these acquisitions ranges from 30° to ∼55°. Values of T_{12} phase larger than 0 correspond to open water, barren land, or areas with low incidence angle.

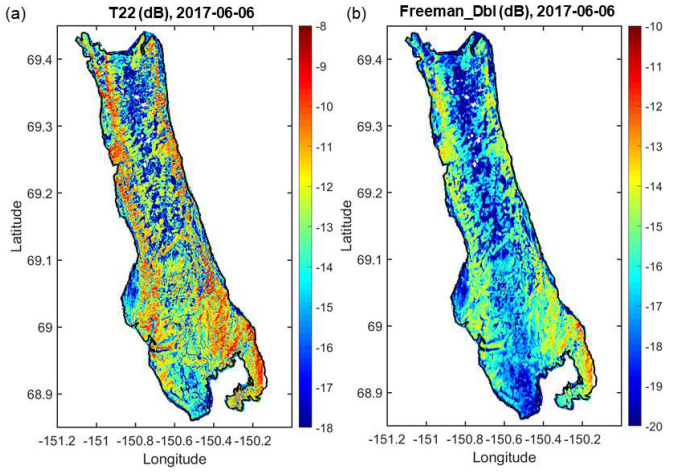


Fig. 9. Maps of the P-band double-bounce scattering power derived from the coherency matrix element (a) T_{22} and (b) Freeman–Durden decomposition within the fire scar in early June 2017. Severely burned areas [Fig. 5(a)] are generally associated with stronger double-bounce scattering power. The majority (∼82%) of the pixels within the burn perimeter have incidence angles ranging from 40° to 55°.

by incidence angle variations and show strong sensitivity to burn severity (Figs. 8 and S8). T_{22} can be interpreted as the intensity of double-bounce scattering, which shows strong increases (>∼4 dB) in the severely burned areas relative to the unburned areas. T_{22} in early June is also generally higher (>2 dB) than in mid-August, likely because wetter surface conditions following spring snowmelt result in stronger double-bounce scattering. T_{22} also shows a similar spatial pattern as the double-bounce power derived from the Freeman–Durden decomposition (Freeman_Dbl) in both periods (Figs. 9 and S9). However, T_{22} is less dependent on the incidence angle based on the regression analysis, as described in Section II-B. Using the linear regression and the observations from the overlapping areas of the two flight lines in June (Fig. S10), T_{22} decreases by 1.09 dB ($r = -0.38$ and $p < 0.1$) with a 10° incidence angle increase, contrasting with a ∼2.3-dB decrease in the Freeman_Dbl power ($r = -0.54$ and $p < 0.1$). However, it should be noted that the two P-band flight lines have limited overlapping areas, and the above regression analysis was conducted only for a

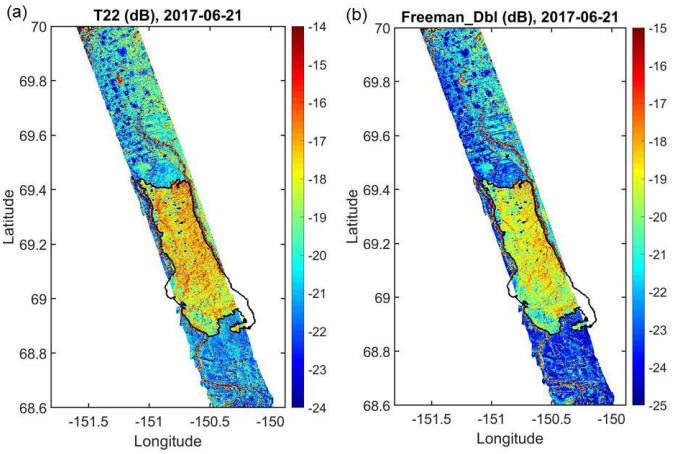


Fig. 10. Maps of the L-band double-bounce scattering power derived from the coherency matrix element (a) T_{22} and (b) Freeman–Durden decomposition in June 21, 2017 (black lines indicating the fire perimeter). Double-bounce scattering was enhanced (>2 dB) in the burned areas. The data were normalized to 50° incidence angle using the overlapping image between the two L-band flight lines.

limited range of incidence angle changes. For comparison, T_{11} (surface scattering power) and T_{33} (volume scattering power) also show large increases in burned areas; however, the power intensity is strongly affected by the local incidence angle, with the mean slopes of 4.4 dB/10° and 3.2 dB/10°.

The T_{12} phase representing the phase difference between double-bounce scattering and surface scattering also increases in the burned areas (Figs. 8 and S8). The T_{12} phase shows a similar pattern as the Touzi dominant-scattering-type phase (TSVM_phi_s1), while a previous study using L-band data showed that the Touzi phase was more sensitive to changes in subsurface water flow of peatlands than the Pauli phase or the HH–VV phase difference [39]. There is more than a 20° difference in the phase angle between the burned and unburned areas, which does not change significantly in different thaw seasons.

As discussed above, most polarimetric parameters, including the surface and volume scattering power, C_{13} phase (HH–VV phase difference), and alpha scattering angle, show a strong dependence on the incidence angle. The majority of pixels within the fire perimeter have incidence angle (INC) ranging from 40° to 55°. Therefore, we conducted the correlation analysis between the polarimetric parameters and the burn severity (represented by the dNBR map) for each INC 5° bin ranging from 40° to 55°. Significant ($p < 0.1$) correlation was found between dNBR with P-band backscatter and most polarimetric variables (Table VI). The correlation is similar for most parameters for June and August images; however, VV backscatter shows a generally lower correlation with dNBR in mid-August than early June. VV backscatter shows a larger decrease in August than HH backscatter due to a general surface drying trend in mid-August, especially in the burned areas, resulting in a strong increase in the HH/VV ratio within the fire scar, which shows better correlation with burn severity in August.

2) L-Band: At L-band, large differences in the scattering power were also observed between the burned and unburned

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TABLE VII
CORRELATION COEFFICIENT BETWEEN THE SCATTERING POWER VARIABLES[1] AND BURN SEVERITY (dNBR) AT L-BAND

	2017-06-21	2017-09-16	2018-08-26
HH (C_{11})	0.42	0.37	0.38
HV (C_{22})	0.32	0.28	0.28
VV (C_{33})	0.28	0.21	0.16
T_{11} ($\|S_{HH}+S_{VV}\|$)	0.37	0.31	0.31
T_{22} ($\|S_{HH}-S_{VV}\|$)	0.32	0.24	0.20
Freeman_Odd	0.44	0.43	0.37
Freeman_Dbl	0.36	0.29	0.27
Freeman_Vol	0.34	0.26	0.24

[1]The scattering power was normalized to 50° incidence angle.

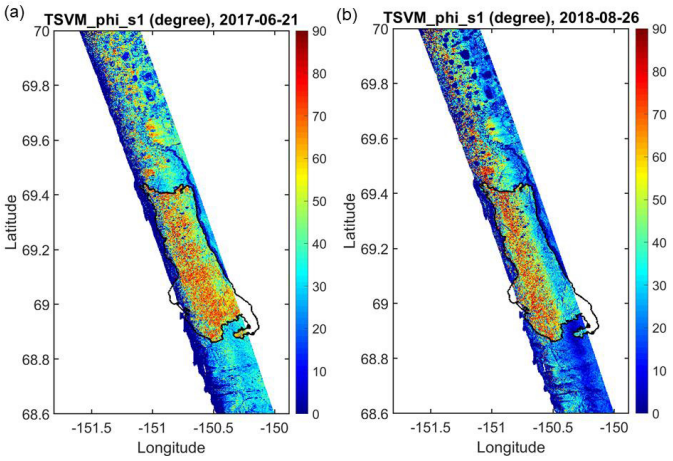


Fig. 11. Comparison of the Touzi dominant scattering-type phase (TSVM_phi_s1) at two different dates derived from the West flight lines (anaktW). The June acquisition shows better contrast between the burned and unburned areas. The incidence angle ranges from 30° to ∼65°.

areas (e.g., Fig. 10). Our results indicate a slightly larger increase in the SAR double-bounce and volume scattering power in the burned areas (>2 dB) than the surface scattering power. The data in June 2017 show the largest differences between the burned and unburned areas, with the smallest differences in August 2018, which corresponds to an overall higher correlation between dNBR and the power variables in June 2017 (Table VII). This is likely due to different surface soil wetness conditions at the three periods. The August 2018 acquisition coincided with minimal accumulated precipitation (<3 mm) during the prior week based on the Daymet data. The SAR surface and double-bounce scattering power are more sensitive to changes in surface wetness than the volume scattering power. The power increases in the burned areas are stronger at HH than at VV polarization, which leads to an overall strong correlation between the co-polarization ratio (HH/VV) with dNBR (Table SIII). However, most L-band variables showed weaker correlation with burn severity compared with the P-band data, which is likely due to saturation of the L-band backscatter intensity in the burned areas as discussed next. In addition, we also found large differences between the power variables and dNBR for different landforms (Table SIV). Most power variables show strong correlation ($r > 0.4$ and $p < 0.1$) with dNBR in the yedoma and glaciated upland areas, while the relatively weak correlation was found in the drained lake basin and rocky upland areas, and nonsignificant negative correlation in the river floodplain areas.

Most phase variables at L-band did not show a significant correlation with burn severity (Table SIII). However, we still found a large contrast in the T_{12} and Touzi dominant scattering phase (TSVM_phi_s1) between burned and unburned areas for the June 2017 acquisition (Fig. 11). T_{12} phase is significantly correlated with dNBR in June ($p < 0.1$). However, the phase angle is very sensitive to incidence angle changes, and overall smaller differences were observed in the phase angle between the burned and unburned areas at very large or low incidence angle.

3) P- and L-Band Comparison: Fig. 12 shows the probability density function of scattering power at multiple P- and L-band acquisitions. The L-band scattering power generally shows a narrower range than P-band and overall smaller differences between burned and unburned areas, especially for the double-bounce scattering. The power of the three scattering components derived from the Freeman–Durden decomposition shows a similar pattern (not shown). The varying incidence angle at P-band used for the comparison should play a minor role in the power difference between P- and L-bands, especially for T_{22}. As we discussed before, T_{22} does not show strong dependence on the incidence angle (<0.1 dB with 10° changes in the incidence angle). Therefore, the difference between the P- and L-bands may be mainly due to the saturation of backscattering power at L-band, which also results in lower L-band backscatter correlation with dNBR (Table VII). For the P-band, during the spring thaw period (June 6, 2017), the double-bounce power shows the largest contrast between burned and unburned areas (mean difference ≈ 3.4 dB), followed by volume scattering power (2.8 dB), with a smaller difference in the surface bounce power (1.7 dB). In mid-August, all scattering power decreased, with slight changes (∼0.2 dB) in the differences of the surface and volume scattering power between the burned and unburned areas. The L-band scattering power also showed a slightly larger contrast between the burned and unburned areas (mean difference ≈ 2 dB) during the spring thaw period (June 21, 2017), with the smallest contrast in August 26, 2018. The backscatter differences between the burned and unburned areas decreased by ∼0.2 dB for volume scattering and ∼0.5 dB for double-bounce scattering and surface scattering in August 2018, compared to June 21, 2017.

Fig. S11 compares the probability density function of Touzi dominant scattering phase at P- and L-band acquisitions. The L-band Touzi phase angle shows a larger dynamic range than P-band, but with smaller mean differences (12°–15°) between the burned and unburned areas compared with P-band (∼25°). This is consistent with overall higher correlation between

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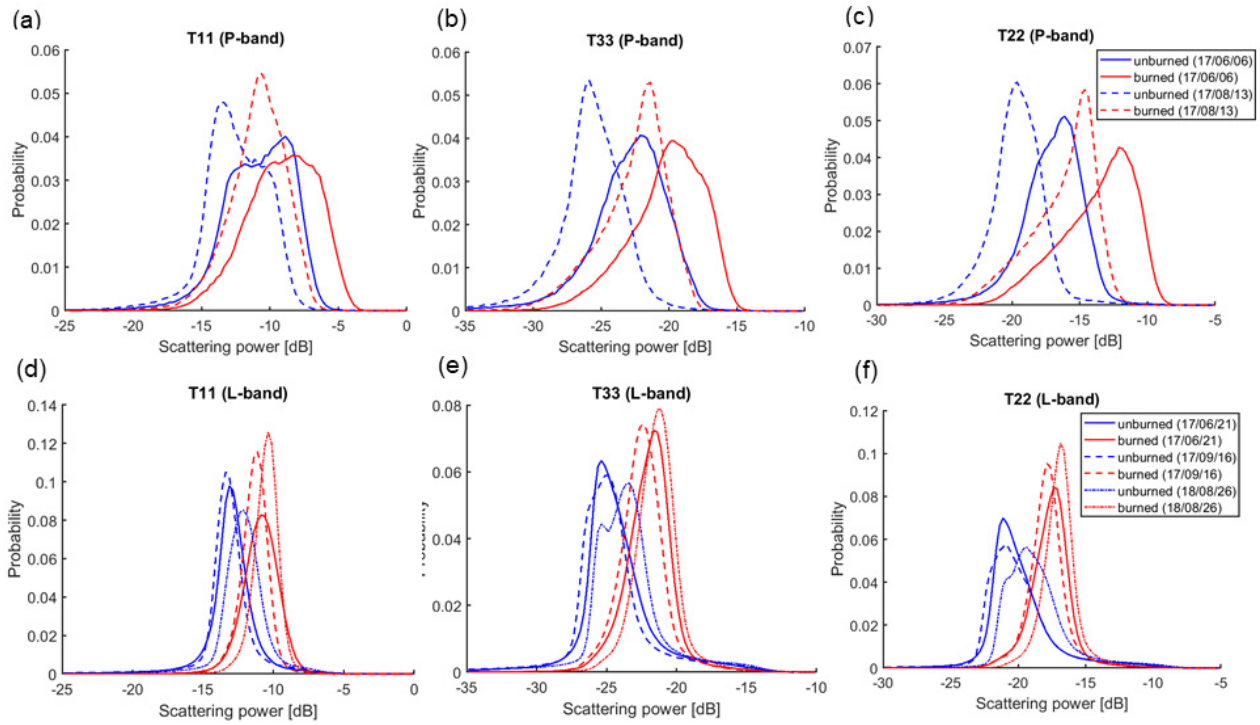


Fig. 12. Comparisons of the histogram of surface bounce (T_{11}), volume (T_{33}), and double-bounce (T_{22}) scattering power between burned and unburned areas represented by the coherency matrix elements from P- and L-band acquisitions. Scattering power for all types was enhanced in the burned areas, especially for the volume and double-bounce scattering. The comparison is conducted for flight transects between 68.75°N and 69.8°N, which have approximately the same coverage for unburned and burned areas. For the P-band, all tundra pixels with incidence angle from 40° to 60° (mean incidence angle ≈ 50°) were included, while the L-band scattering power was normalized to 50° incidence angle.

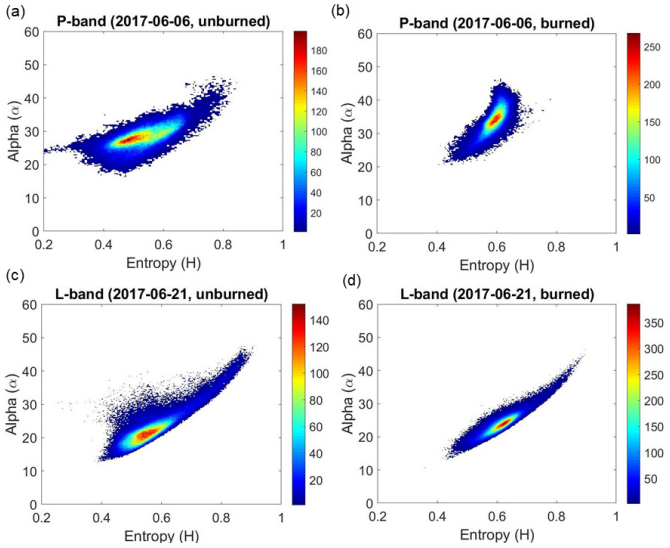


Fig. 13. 2-D entropy-alpha plane for unburned and burned areas derived from (a) and (b) P-band and (c) and (d) L-band data with incidence angles from 45° to 55° in June 2017, for flight transects between 68.75°N and 69.8°N (same as Fig. 12). Barren and surface open water areas were excluded. Burned areas show an overall larger scattering angle and increased randomness at P-band, while nonsignificant differences are observed at L-band.

dNBR and the P-band Touzi phase angle. The P-band data during the spring thaw period can slightly better distinguish the burned and unburned areas with a mean phase difference of 26°, but the difference between spring and later thaw period is not significantly different.

The Cloude–Pottier decomposition results show a slight increase in the surface scattering angle (alpha) and more randomness (entropy) of the surface scattering process in the burned areas at both P- and L-bands, but with slightly larger P-band contrast between the burned and unburned areas (Fig. 13). This is likely due to increasing surface roughness and microtopography associated with fire-induced ice-wedge degradation and thermokarst development (Fig. 2). The L-band generally shows a larger dynamic range in both scattering angle and entropy values than the P-band. It is also interesting to see that the P-band generally shows slightly larger scattering angle than the L-band, which may be due to differences in the acquisition timing of the two flights. The P-band acquisition occurred at the start of the spring thaw, which can result in very wet surface soil conditions and enhanced double-bounce scattering. There was almost no precipitation during June 2017 (indicated by the Daymet data). Thus, with gradual drying, surface scattering is likely enhanced and double bounce reduced by late June when the L-band data were acquired.

IV. Discussion

Our results highlight the potential of PolSAR data, especially at the longer wavelengths such as P-band, in characterizing postfire changes in surface scattering processes at decadal time scale, due to fire-induced changes in surface and subsurface processes in the permafrost region, including thermokarst formation, removal of surface vegetation and organic layer, and active-layer deepening and potential drying trends. As shown in Fig. 14, fire-induced ice-wedge

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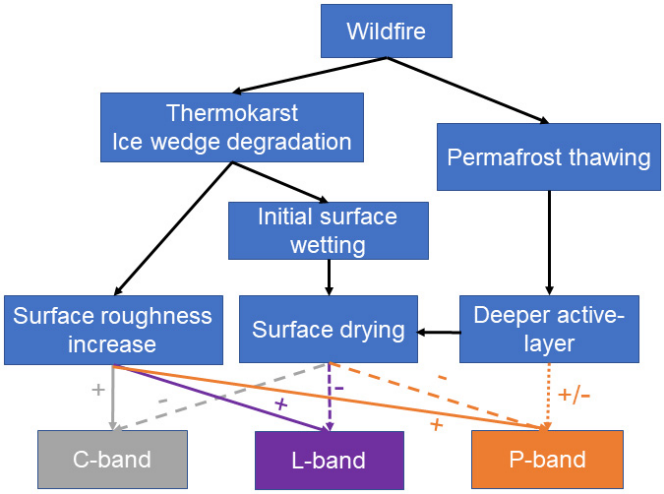


Fig. 14. Conceptual diagram showing the potential impact of wildfire on the permafrost processes in the Arctic tundra region, which may enhance (+) or reduce (-) the radar backscatter. The postfire vegetation recovery can also have a large impact on the radar backscatter response, especially at the higher frequency; however, this is not included in the diagram.

degradation and thermokarst formation can lead to increasing surface roughness and surface drying at a longer time scale. Removal or reduction of the surface organic layer will reduce the surface insulation effects on the underlying permafrost and promote active-layer thawing, which can lead to a deeper and drier active-layer especially at the surface. All these changes can contribute to the observed backscatter differences between burned and unburned areas in the ARF region. Moreover, such differences can vary seasonally due to the sensitivity of radar backscatter to different factors during the frozen and thaw season as discussed next.

A. Response of Multifrequency PolSAR Data to Postfire Permafrost Changes and Recovery Processes

Our results show consistently higher correlation of C-band winter backscatter with burn severity than the summer backscatter. This is because winter backscatter is less affected by variations in the surface soil moisture and vegetation water content, but more by surface roughness. However, snow cover and underlying frozen soil conditions may also affect the C-band winter backscatter [40]. In 2017, the Daymet data show a general uniform and shallow snow cover in the ARF region with mean snow water equivalent (SWE) of ∼30 mm in March from north to south (Fig. S12). The winter backscatter across the sites showed consistent differences among different landforms (Fig. 4), which showed characteristically different increasing surface microtopography based on the LiDAR survey in 2009 and 2014 (Fig. S13) [5]. We found overall higher correlation between the microtopography changes derived from the above LiDAR survey and C-band backscatter during the winter period ($R = 0.22 \pm 0.02$) than the summer period ($R = 0.01 \pm 0.03$). Among all the landforms, regions with larger changes in surface microtopography, such as the yedoma and glaciated upland regions, show higher correlation ($R > 0.25$ and $p < 0.05$) with winter backscatter than other regions. Therefore, we assume that the variations in the C-band winter backscatter in 2017 are mainly affected by changing surface roughness. This was also partially supported by higher correlation between the C-band winter backscatter in 2017 and dNBR for the yedoma region with greatly increased surface roughness during postfire [5] (Table IV). However, we did not find a strong correlation between the C-band winter backscatter and dNBR in 2018 as in 2017. The spatial pattern in the 2018 winter backscatter seems more consistent with the spatial variations of the snowpack (Figs. S12 and S14). In 2018, the southern part of the domain (∼< 69.05 °N) has much higher SWE (∼>50 mm) than the north (∼30 mm), while the winter backscatter is ∼2 dB lower in the south. Changes in the scattering processes due to differences in the snowpack and near-surface frozen soil conditions may explain the north–south gradient in the C-band winter backscatter in 2018 (Fig. S14).

Few studies have discussed the fire impact on the winter radar backscatter in the boreal and Arctic region [20], [41], [42], which can differ significantly between the tundra and boreal forest, due to variations in the snow cover, canopy structure, and ground freeze–thaw state conditions. Focusing on the ARF region, Zhou *et al.* [20] found virtually no difference in the winter backscatter between the burned and unburned areas for both L- and C-bands during the first two postfire years; after that, the winter backscatter in the burned areas remained ∼2 and ∼1 dB higher for L-band and C-band, respectively. In contrast, fire scars in boreal forest systems exhibited distinctly different postfire backscatter behavior. A study of a forest-tundra fire scar in the Chukotka region of eastern Siberia [41] indicated that the presence of snow cover smoothed the contrast in the C-band winter backscatter between the burned and unburned areas; however, the winter backscatter in the burned areas was still ∼2 dB higher than the unburned areas during the first two years after fire. A study in the Alaskan boreal forest [42] showed that the C-band backscatter was >4 dB lower in the burned areas during the first 1–2 years after fire, likely due to different impact of ice/snow conditions in the burned and unburned areas.

There was no available winter coverage of airborne P- or L-band data, so we only compared the radar responses at different frequencies during the thaw season. Overall, our results show much larger radar backscatter differences at longer wavelengths (particularly the P-band) between the burned and unburned areas than for C-band. With gradual vegetation recovery, we observed limited differences in the C-band summer backscatter between the burned and unburned areas at ten-year postfire, which is consistent with previous studies [18]–[20]. We did not notice a close association between the C-band summer backscatter and burn severity either. Different soil moisture trends associated with burn severity are expected to occur in different landforms, resulting in a divergent response of summer backscatter across the domain (Fig. S3). However, the SAR data at longer wavelengths have deeper penetration depths and are more sensitive to potential fire-induced changes in the surface roughness and subsurface soil processes, which may take much longer to recover than the surface vegetation cover [11], [12]. Previous studies have also

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showed a longer recovery of L-band backscatter in boreal forest and tundra [18], [20]. In our study area, increasing surface roughness in severely burned areas enhances all components of the P- and L-band scattering power, but with stronger increase in the double-bounce scattering and volume scattering than surface scattering. At P-band, the severely burned areas show $>\sim$3-dB increase in the double-bounce power during the early spring thaw period when the surface is very wet due to snowmelt water pooling. For both P- and L-bands, the HH polarization shows better correlation with burn severity and a large contrast in the HH backscatter between the burned and unburned areas was observed under all surface conditions, whereas VV backscatter is more sensitive to surface scattering, and shows large decrease when the surface is drier during the later thaw season. Previous studies have reported that the HH/VV ratio and phase at L-band are sensitive to the subsurface soil profile [43]–[45], which is likely due to the different sensitivity of HH and VV backscatter to subsurface scattering. Further understanding of this sensitivity may improve our capabilities for regional monitoring and modeling of soil active-layer dielectric and moisture profiles during postfire recovery in permafrost regions [46].

P-band shows consistently larger contrast in backscatter power and phase between the burned and unburned areas compared to L- and C-bands, which makes it potentially more useful to study fire–permafrost interactions in the Arctic over decadal time scales. The P-band data show a larger dynamic range of backscatter intensity in responding to burn severity than the L-band data even after ten years of postfire recovery (Fig. 12). The P-band radar responses to fire also vary between the early and the late thaw season, while the L-band data show a smaller backscatter seasonal and disturbance response. For example, both the alpha angle and the randomness of scattering at P-band decrease at the later thaw period relative to the early thaw period, while the L-band scattering parameters are rather similar over the three acquisition dates. We hypothesize this to reflect stronger P-band sensitivity to the subsurface soil moisture and dielectric profile [30], [34] and a relatively slow postfire recovery of the tundra soil active layer [11]. Wetter soil conditions in the Arctic permafrost region generally limit the penetration depth of L-band to a few centimeters. However, inconsistent timing of the two flight acquisitions adds uncertainty to our conclusion. The P-band data were acquired in early June during the early spring thaw period following seasonal snowmelt when the land surface is relatively wet and double-bounce scattering plays an important role. The L-band data were acquired 15 days later under drier surface conditions, which may contribute to differences in the dominant scattering processes observed at the two frequencies. Ideally, multifrequency acquisitions from the same flight as the DLR's new airborne SAR system F-SAR are desired.

B. Potential Impact of Fire, Vegetation, and Permafrost Interactions on the Radar Response

The distinct differences in the radar response between burned and unburned areas and between different landforms should be mainly caused by fire-induced permafrost changes, including changes in the active-layer thickness, surface wetness and soil moisture, microtopography, and surface roughness, as shown in Fig. 14.

Field surveys since the fire occurrence in 2007 show a gradual recovery of surface vegetation in terms of both diversity and density, as well as the surface organic layer [26]. The most recent field survey in 2017 showed deciduous shrub cover at burned transects (22.3% $\pm$ 16.3%) had recovered to unburned reference transect levels (28.2% $\pm$ 17.3%), but not the ericaceous shrub cover (burned: 17.9% $\pm$ 11.6% versus unburned: 42.4% $\pm$ 18.6%). The shrub counts were slightly greater for deciduous species in the burned areas (5.48 $\pm$ 4.31 m^{-2}) than the unburned areas (3.48 $\pm$ 1.69 m^{-2}); however, the shrub cover is likely a better indicator of shrub density in our study area. We did not find a significant correlation between the C- or normalized L-band backscatter and shrub cover, particularly the ericaceous shrub cover (absolute correlation $<\sim$0.1). Although the C-band VV backscatter during winter (relative to the summer period) shows a higher correlation with the cover of deciduous shrubs and subshrubs ($R = -0.30 \pm 0.12$ and $N = 25$), the correlation is still nonsignificant ($p > 0.05$). For the L-band, the cover of deciduous shrubs and subshrubs shows a higher correlation with HH backscatter ($R \sim = -0.38$, $N = 16$, and $p > 0.05$), comparing with much lower correlation with VV or VH backscatter. However, most unburned sites were outside of the L-band image, which may affect the correlation analysis. Therefore, the differences in the shrub regrowth are likely minor factors contributing to the backscatter difference between the burned and unburned areas. This is particularly true for the longer wavelengths (e.g., P-band and L-band) with deeper penetration. However, limited field measurements may introduce large uncertainties to the above analysis.

Besides surface vegetation, fire also consumes a large amount of surface organic matter [1]. The field survey in 2017 showed that the surface organic layer was accumulating at a rate of $\sim$0.47 cm/year, and the mean organic soil depth ($\sim$20.8 cm) at burned sites is slightly shallower than the unburned sites ($\sim$21.5 cm). Reduced organic layer thickness and surface vegetation cover and decreasing surface albedo during the initial postfire period can substantially warm the soil and increase the active layer depth [28]. Our records also indicated that the soil temperatures at 15 cm depth in the burned areas were 1.5 °C warmer on average than in the unburned sites [26]. Previous studies also showed that the soil active-layer recovery in permafrost tundra can take decades [11].

Widespread fire-induced thermokarst has been observed in the ARF burn area, particularly in the ice-rich permafrost region, such as the yedoma upland areas (Fig. 2). For example, Jones *et al.* [5] analyzed airborne LiDAR data acquired in 2009 and 2014 within the ARF perimeter and reported large increase in surface microtopography and roughness in the yedoma areas (Fig. S13) as a result of ice-wedge degradation. Moderate changes were also observed in the rocky and glaciated uplands, followed by the drained lake basins, while negative changes were observed in the river floodplain area. This is consistent with our correlation analysis of radar backscatter with both burn severity and LiDAR-derived surface

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microtopography changes from 2009 to 2014. Highest correlations between radar backscatter and burn severity or microtopography changes were observed in the yedoma and glaciated upland regions, with reduced correlation in the drained basin areas, and lowest correlation in the river floodplain areas (Tables SIV and SV). Larger L-band backscatter differences between burned and unburned sites were also observed in the yedoma and glaciated upland areas than in the drained basin areas (Figs. 7, S6, and S7). Thermokarst formation can also lead to changes in the surface and subsurface hydrology; the soil tends to become wetter due to surface inundation during the initial stages of thermokarst formation, followed by a gradual drying trend due to increasing surface drainage [47]. Therefore, a complex pattern of landscape drying and wetting may be observed in the fire-affected Arctic region due to active-layer deepening, thermokarst formation, and subsequent changes in the soil drainage pattern, especially during the early postfire period. With continued warming, the Arctic region is likely experiencing more frequent wildfire [7]. The interactions of fire disturbance, vegetation, and permafrost alter the surface and subsurface hydrology in ways that are spatially complex and may be beneath the effective resolution of many global satellite sensors and Earth system models [46], [47]. However, our results show that the low-frequency (L- and P-bands) airborne SAR retrievals are sensitive to postfire changes in tundra surface and active layer conditions, capturing relatively fine-scale heterogeneity in the long-term hydrogeomorphic changes that may be missing from the coarser scale satellite observations and model simulations.

V. Conclusion

Our results showed strong sensitivity of L- and P-band PolSAR data to fire-induced changes in the surface and subsurface soil processes in the Arctic permafrost region, relative to C-band data. A large increase in the radar backscatter intensity and phase was observed between the burned and unburned areas from both P- and L-band data at ten-year postfire. An increase in radar backscatter from severely burned areas can exceed ∼4 dB at both frequencies. Enhanced double-bounce scattering and volume scattering ($>\sim$2–3 dB) were also observed, while there were relatively small increases in surface-bounce scattering power. Backscatter intensity and phase at P-band show overall stronger correlation with burn severity than at L-band, which is likely due to L-band saturation to burn severity. Even though limited differences were found in the C-band backscatter between burned and unburned areas during the thaw season, the burned areas showed overall enhanced backscatter during winter ($\sim>1$ dB). The fire-affected ice-rich yedoma areas showed the largest backscatter increases and the highest correlation with burn severity at both C- and L-bands. Increasing surface roughness and microtopography, associated with fire-induced thermokarst development and ice-wedge degradation, should contribute to overall higher backscatter intensity and more random scattering in the burned areas. In addition, an overall drier and deeper active layer can also increase signal contributions from subsurface scattering, which may enhance double-bounce and volume scattering effects, particularly for P-band.

Our results highlighted the potential of longer wavelength full PolSAR data (especially P-band) in characterizing decadal postfire changes in the surface and subsurface soil processes due to fire-induced permafrost degradation in the Arctic tundra. A better understanding of the SAR scattering processes at different frequencies is expected to improve remote sensing capabilities for regional monitoring and the representation of fire–permafrost interactions in physical process models.

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